A Hitchhiker’s Guide to Causal Inference

Guest Speaker

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Welcome

Session Goals

Causal Inference – Core Ideas

- Potential outcomes framework
- DAGs and Simpson's paradox
- Correlation and causal bias

Estimating Causal Effects – Average Treatment Effect

- Matching Methods
- Inverse Propensity Score Weighting (IPW)
- Double Machine Learning

Industry Applications

- Casual Discovery
- Surrogate methods for marketing measurement
- Targeted Maximum Likelihood Estimation

Causality

Terminology

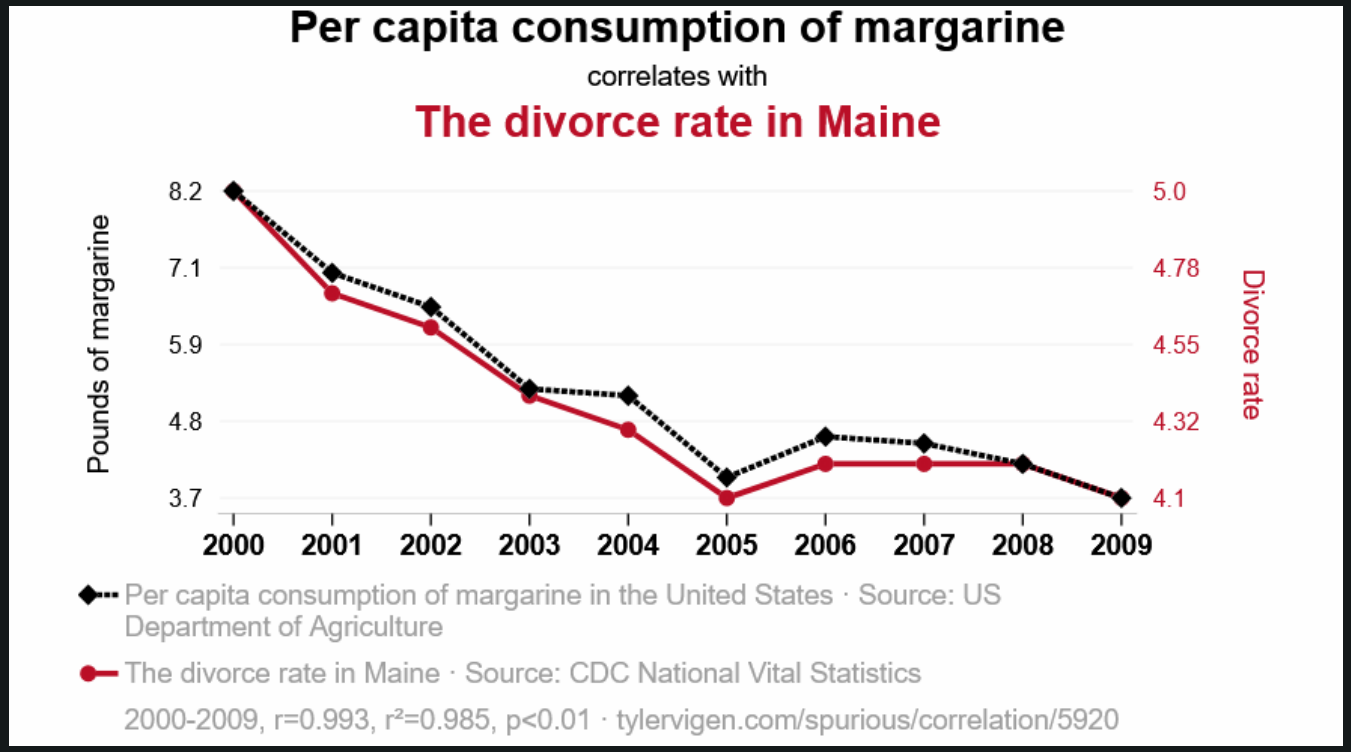
X causes *Y* if, were we to intervene and change the value of *X*, then the distribution of *Y* changes as a result

Directed Acyclic Graphs (DAGs)

Association

Correlation is helpful for prediction, can mislead on causation 😬

Does the dairy aisle hold the key to marital bliss?



Association

Many ML algorithms model associations

$$E(Y|X) = f(X) + \varepsilon$$

Hotel Bookings

- **Predict** occupancy rate from room price and other features
- Goal – Maximize occupancy all year
- ML model might say … "*increase prices*"

Customer Targeting

- Predict customer conversion from activity, spending, …
- Goal – Maximize conversions and minimize marketing costs
- ML model might say … "*Target the most active customers*"

Core Idea

Predicted occupancy is100% when room rates are $450 per night. If I intervene and change my rates from $150 to $450, will all my rooms fill up this week?

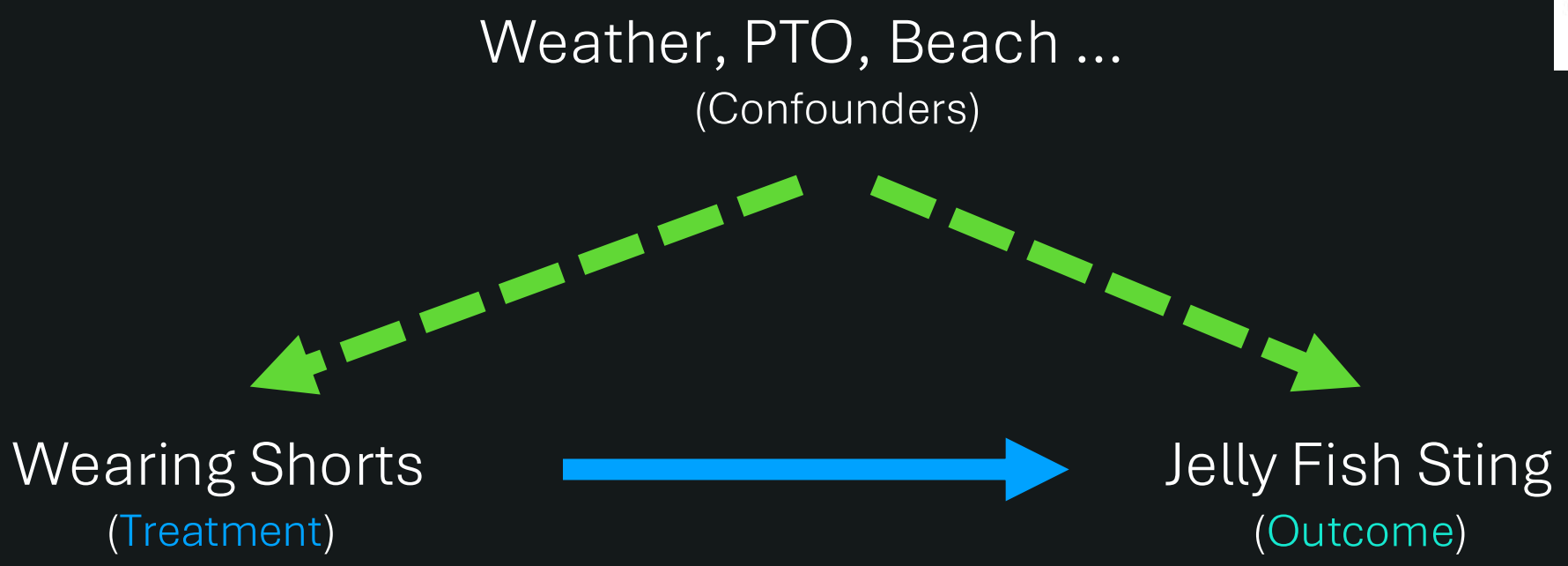
$$E(Y|X = 450) - E(Y|X = 150)$$

Association Without Causation

Confounding

Does Wearing Shorts (Treatment) have a causal effect on Jelly Fish Stings (Outcome)?

- *Yes*, pants lower your chances of being forced into a game of "Sting Pong" on your bare ankle
- Association in observational data is mostly driven by confounding



Simpson's Paradox

When results change direction within subgroups of the population

Does using treatment A or B work better for reducing mortality of a disease?

Treatment	Total
A	16% (240/1500)
B	19% (105/550)

Treatment	Condition: Mild	Condition: Severe	Condition: Total
A	15% (210/1400)	30% (30/100)	16% (240/1500)
B	10% (5/50)	20% (100/500)	19% (105/550)

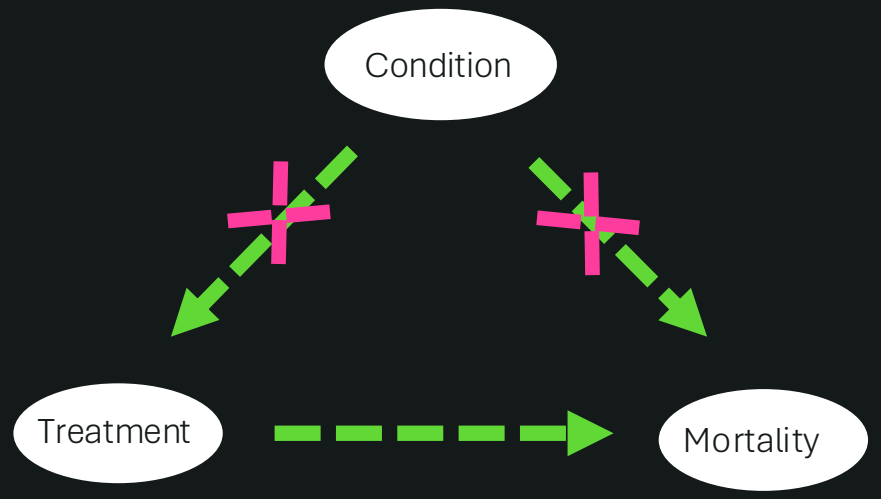
Simpson's Paradox

DAGs (Directed Acyclic Graphs) to the rescue

Disease severity has a causal impact on both treatment and mortality

Using treatment B is better in this world

		Condition		
		Mild	Severe	Total
Treatment	A	15% (210/1400)	30% (30/100)	16% (240/1500)
	B	10% (5/50)	20% (100/500)	19% (105/550)



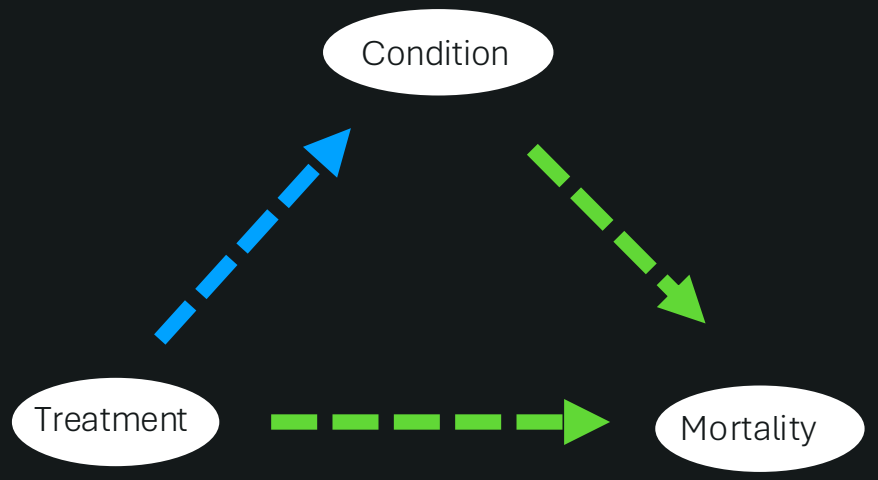
Simpson's Paradox

Which treatment is better? Another possibility

Treatment has a causal impact on disease severity (condition) and mortality

Using treatment A is better in this world

		Condition		
		Mild	Severe	Total
Treatment	A	15% (210/1400)	30% (30/100)	16% (240/1500)
	B	10% (5/50)	20% (100/500)	19% (105/550)



Potential Outcomes Framework

Terminology

X – Set of confounding variables

Customer attributes and behaviors

T – Treatment variable

Typically, a binary intervention (0/1)

Y^0

Outcome if customer does not receive treatment

Y^1

Outcome if the same customer does receive treatment

ATE

Average treatment effect

$$E(Y^1 - Y^0)$$

Example

Treatment (T) – Show Advertisement

Outcome (Y) – Signup for a Service

	X		Y^1	Y^0	Individual Treatment Effect (ITE)
customer	music_streams	is_retail	Potential Outcomes		Y^1 - Y^0
1	24	1	1	0	1
2	2	0	0	0	0
3	120	1	1	0	1
4	342	1	1	1	0
5	53	0	0	1	-1
			Average Treatment Effect (ATE)		0.20

Potential Outcomes

The Real World 😥

X – Confounding variables

T – Treatment variable

Y - Observed outcome

	X		T	Y	Y^1	Y^0
customer	music_streams	is_retail	shown_ad	sign_up	Potential Outcomes	
1	24	1	1	1	1	0
2	2	0	0	0	0	0
3	120	1	1	1	1	0
4	342	1	1	1	1	1
5	53	0	0	1	0	1
	Average Treatment Effect?			0.5		

Estimating ATE

$E(Y|T = 1) - E(Y|T = 0)$???

Fundamental Problem of Causal Inference

Hands-On Coding

Potential Outcomes in Python

Causal Assumptions

Assumptions shared across modeling methodology

Consistency

- The outcome seen in assigned treatment is the same as the corresponding potential outcome

No unmeasured confounders (Exchangeability)

- There are no hidden variables that are driving your treatment and outcome which are unaccounted for, i.e. “No Hidden Back Doors”

Positivity

- The chance of receiving treatment is non-zero for all combinations of covariates/confounder variables

Potential Outcomes

Causal Estimation Bias

$$E(Y|T=1) - E(Y|T=0)$$

$$= E(Y^1|T=1) - E(Y^0|T=0)$$ *Consistency Assumption*

$$= E(Y^1|T=1) - E(Y^0|T=0) + \mathrm{E}(Y^0|T=1) - \mathrm{E}(Y^0|T=1)$$

$$= E(Y^1|T=1) - \mathrm{E}(Y^0|T=1) + \{\mathrm{E}(Y^0|T=1) - E(Y^0|T=0)\}$$

$$= E(Y^1 - Y^0|T=1) + \{\mathrm{E}(Y^0|T=1) - E(Y^0|T=0)\}$$

= ATE (Treatment Group) + Bias

Bias in Causal Inference

Marketing Example

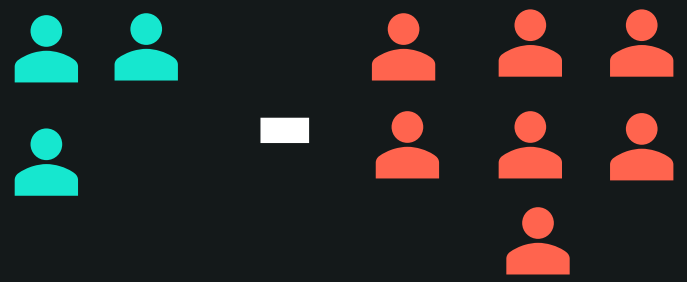
Y - Customer converted (Yes/No)

T – 5+ marketing messages in last 30 days (Yes/No)

X – Engagement, plan type, devices, retail, …

Estimating ATE

- $E(Y|T=1) - E(Y|T=0)$
  - Typically, highly biased due to treatment assignment
- Main source of bias is $E(Y^0|T=1)$ vs $E(Y^0|T=0)$
- What can we do?
  - Randomized Control Trials (RTCs)
  - Causal Modeling
  - Give up and assume 42 is the answer 🤷



15% - 5% = 10% ATE (?)

Hands-On Coding

Modeling with Scikit-Learn in Python

Matching Methods

Addressing Confounding Bias

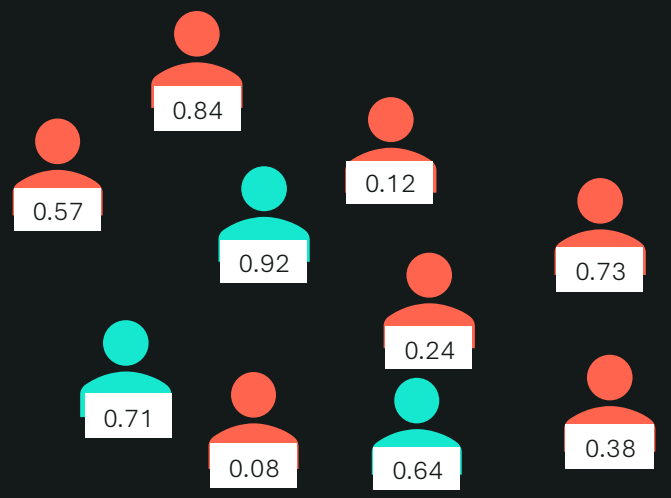
Y - Customer converted (Yes/No)

T – 5+ marketing messages in last 30 days (Yes/No)

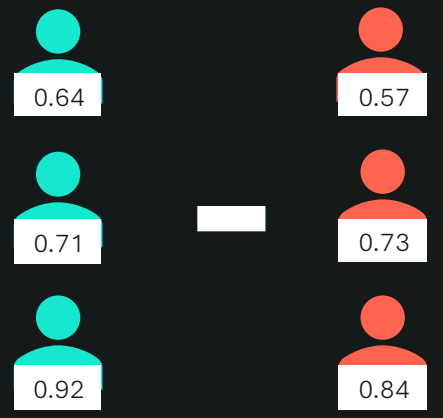
X – Engagement, plan type, devices, retail, ...

ML Model predicts T using X

Provides likelihood to receive treatment



Match customers on likelihood to receive treatment and calculate the average difference, mimicking an RCT



9% - 4% = 5% ATE

Matching Methods

Inverse Propensity Score Estimation

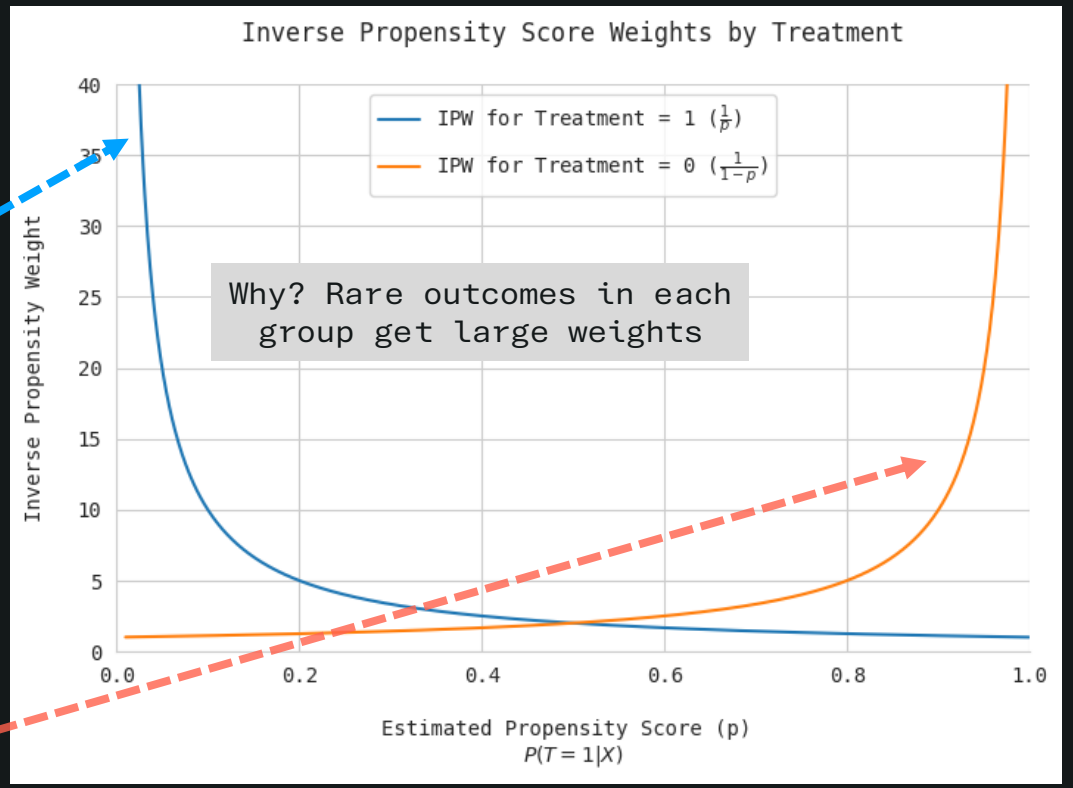
Balancing treatment groups with *propensity scores*

1. Estimate $p_i = P(T = 1 \mid X_i)$ (typically Logistic Regression)

2. Calculate weight for each observation

For T = 1: $w_i = \frac{1}{p_i}$

For T = 0: $w_i = \frac{1}{1 - p_i}$



Matching Methods

Inverse Propensity Score Estimation

Balancing treatment groups with *propensity scores*

For T = 1: $w_i = \frac{1}{p_i}$ For T = 0: $w_i = \frac{1}{1 - p_i}$

3. Calculate weighted average within each treatment group to estimate $E(Y^1)$ and $E(Y^0)$

$$ATE = E(Y^1) - E(Y^0)$$

$$\frac{\sum_i Y_i(\frac{1}{p_i})}{\sum_i \frac{1}{p_i}} - \frac{\sum_i Y_i(\frac{1}{1 - p_i})}{\sum_i \frac{1}{1 - p_i}}$$

Matching Methods

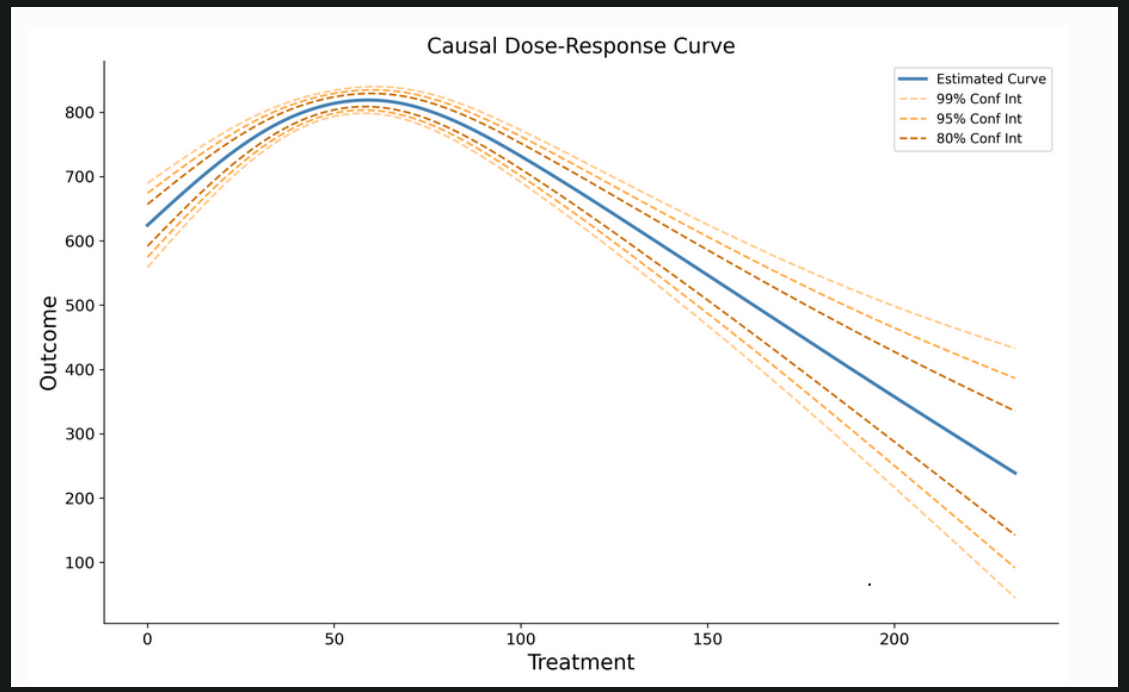
Generalized Propensity Score Matching (GPS)

GPS extends matching methodology to non-binary treatments

- Continuous – milliliters (ml) of drug administered
- Count - Number of digital ads seen in the last month

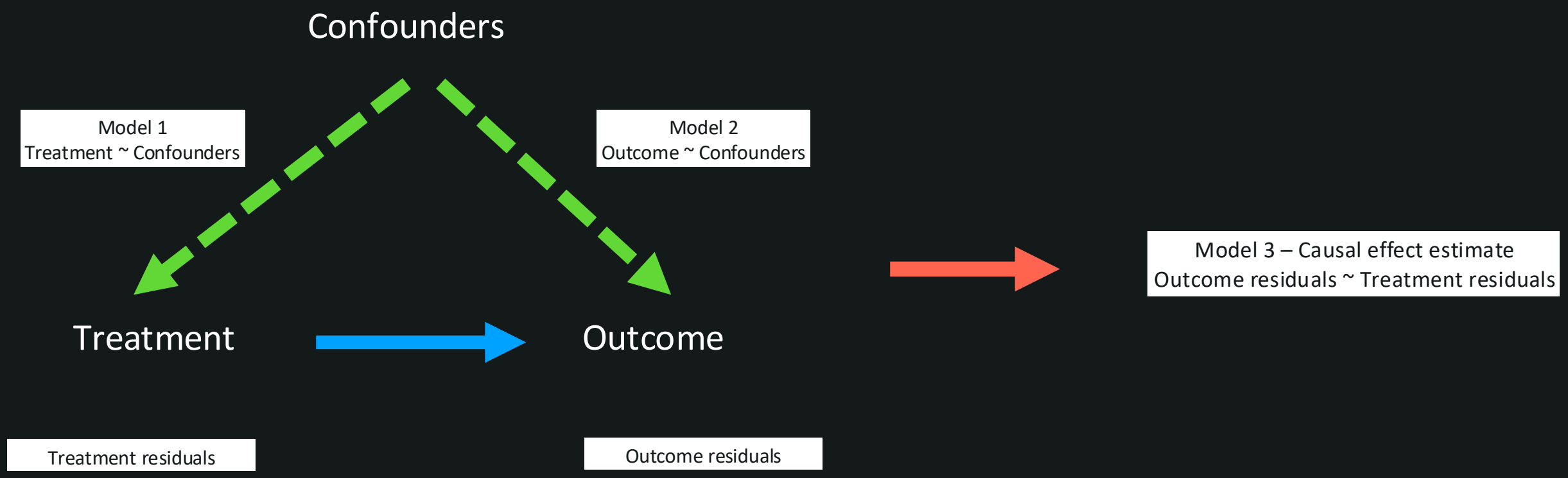
Rooted in biostatistics and clinical trials

- Dose-Response curve



Double Machine Learning

Modeling the outcome and treatment



Hands-On Coding

Estimating ATE with Matching and Double ML in Python

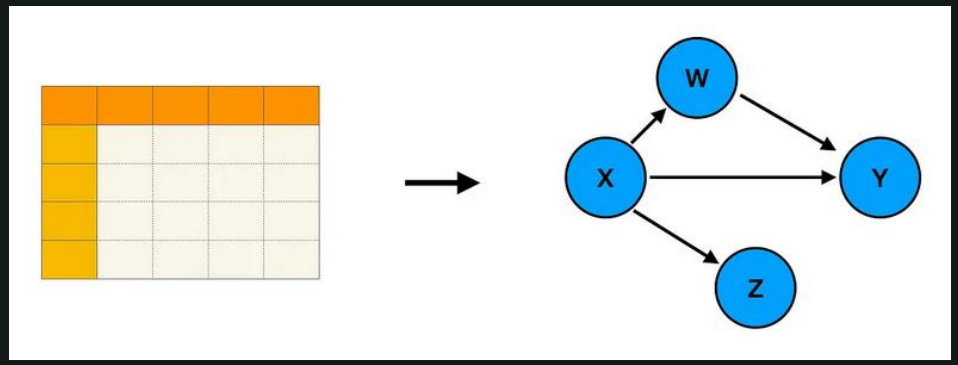
Industry Applications

Causal Discovery Algorithms

Mining the causal structure of data

The goal of causal discovery is to infer causal structure from data

- Given a dataset, find a set of potential causal diagrams that describe the data generating process (DGP)



Lots of exciting developments in the Python community

causaLens
Causal AI: Causal Graphs

Causal Discovery Toolbox

Experimentation

Surrogate modeling for business impact estimation

1. Causal Discovery to find customer high value actions
2. DML to estimate causal impact on long-term business outcome
3. Link changes to HVA activity during experiments to long-term business impact

```
dml_data = (
    DoubleMLData(
        data=observational_df,
        y_col='converted',
        d_cols='upsell_marketing',
        x_cols=customer_features,
        use_other_treat_as_covariate=False)
)
```

Set to True

Targeted Maximum Likelihood Estimation

Latest framework for estimating causal effects

Minimal Assumptions

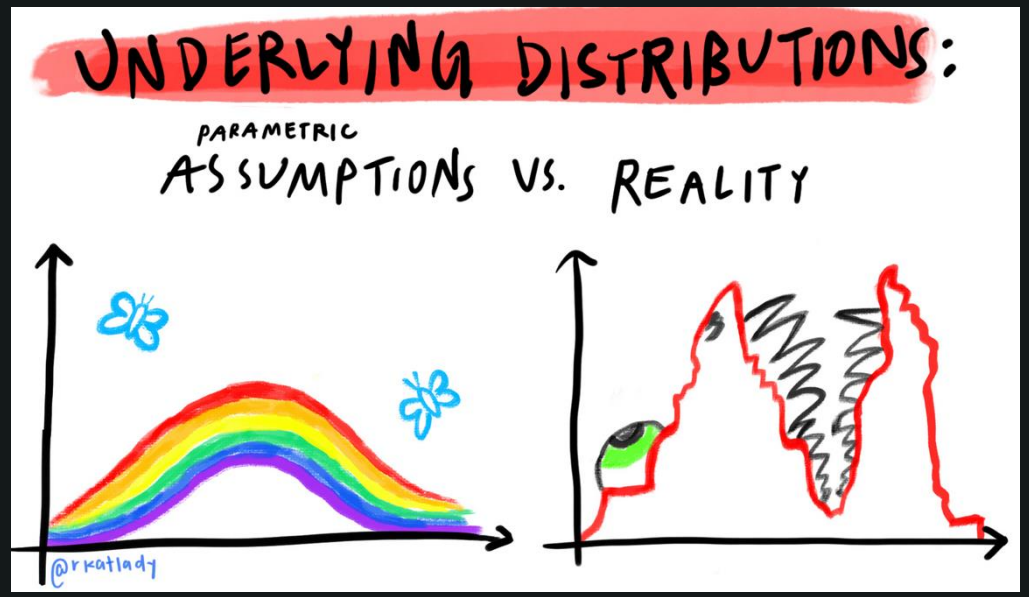
- We have samples from some probability distribution over (Y, T, X)

What We Estimate

- Not a parameter in a statistical model (i.e. $Y \sim \beta_0 + \beta_1 T + \beta_2 X$)
- Instead, we estimate a parameter of the probability distribution

Average Treatment Effect

$$\varphi_0 = E(E[Y|T = 1, X] - E[Y|T = 0, X])$$



Closing Remarks

Causal ML

Closing Remarks

Why Hasn't Causal ML Taken the World by Storm?

- Highly efficient methods for a wide range of estimation problems
- Relatively new in industry ~ 10 years
- It's complicated
  - ML, semi-parametric statistical theory, measure theory 🤯

Artificial General Intelligence

Is AGI coming to get us?

The good news is that since causal ML requires deep understanding of business processes and reasoning about the underlying causal mechanisms - it is safe from the AI takeover ... for now

Thank You